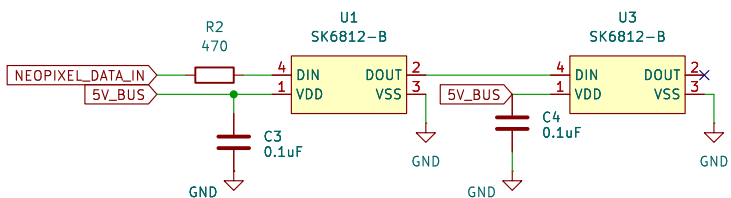
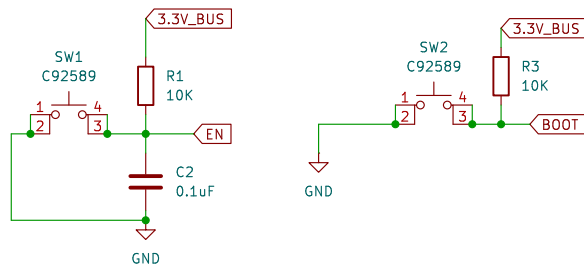


File: H_Bridge.kicad_sch



BOOT BUTTON



The diagram shows the ESP32-S3-WROOM-1U module with the following connections:

- Power:** 3.3V BUS to VDD (pin 37), GND to GND (pin 30).
- Reset:** EN (pin 3) to a reset button.
- Capacitors:** C6 (10uF) and C7 (0.1uF) connected to VDD and GND.
- I2C:** SDA (pin 28) to A4, SCL (pin 29) to A5.
- SPI:** CS (pin 31) to a chip select button, MISO (pin 32) to D0, MOSI (pin 33) to D1, CLK (pin 34) to D2.
- Other Pins:**
 - GPIO0/BOOT (pin 27) to BOOT button.
 - GPIO1 (pin 39) to GPIO1.
 - GPIO2 (pin 38) to GPIO2.
 - GPIO4 (pin 4) to GPIO4.
 - GPIO5 (pin 5) to GPIO5.
 - GPIO6 (pin 6) to GPIO6.
 - GPIO7 (pin 7) to GPIO7.
 - A_SLEEP (pin 12) to A_SLEEP.
 - A_IN1 (pin 17) to A_IN1.
 - A_IN2 (pin 18) to A_IN2.
 - LVL_SHIFTER_IN (pin 19) to LVL_SHIFTER_IN.
 - B_SLEEP (pin 20) to B_SLEEP.
 - B_FAULT (pin 21) to B_FAULT.
 - B_SENSE (pin 22) to B_SENSE.
 - LIPO_VOLTAGE_SENSE (pin 10) to LIPO_VOLTAGE_SENSE.
 - A_SENSE (pin 11) to A_SENSE.
 - B_IN2 (pin 23) to B_IN2.
 - GPIO38 (pin 31) to GPIO38.
 - GPIO45 (pin 26) to GPIO45.
 - GPIO46 (pin 16) to GPIO46.
 - B_IN1 (pin 24) to B_IN1.
 - GPIO48 (pin 25) to GPIO48.

- If you use an "Octal SPI" Module (e.g., N8R8, N16R8):
 - Status: **CRITICAL CONFLICT.**
 - The "R8" versions use 8 data lines to talk to the RAM. GPIO 35, 36, and 37 act as data lines (`SPII05` , `SPII06` , `SPII07`).
 - If you connect CAN to these pins on an R8 module, the chip will crash immediately because you are interfering with its memory access.

Sheet: /
File: TerraDrive.kicad_sch

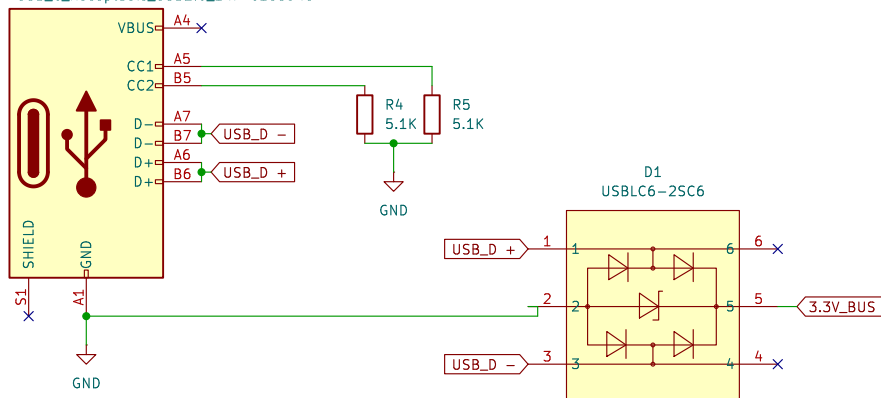
Size: A4	Date:
KiCad E.D.A. 9.0.2-rc1	

Date:

Rev:
Id: 1/4

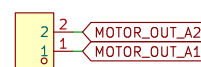


USB_C_Receptacle_USB2.0_14P C165948

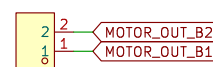


Motor Output Terminals

CN1
KF243-5.0-2P

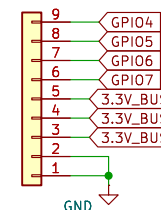


CN3
KF243-5.0-2P



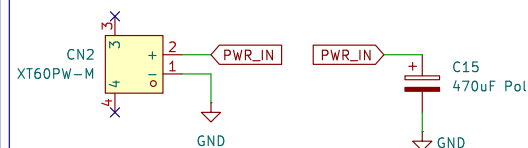
3.3V Logic IO Pins

J4
Conn_01x09

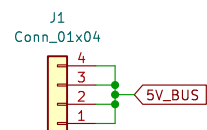
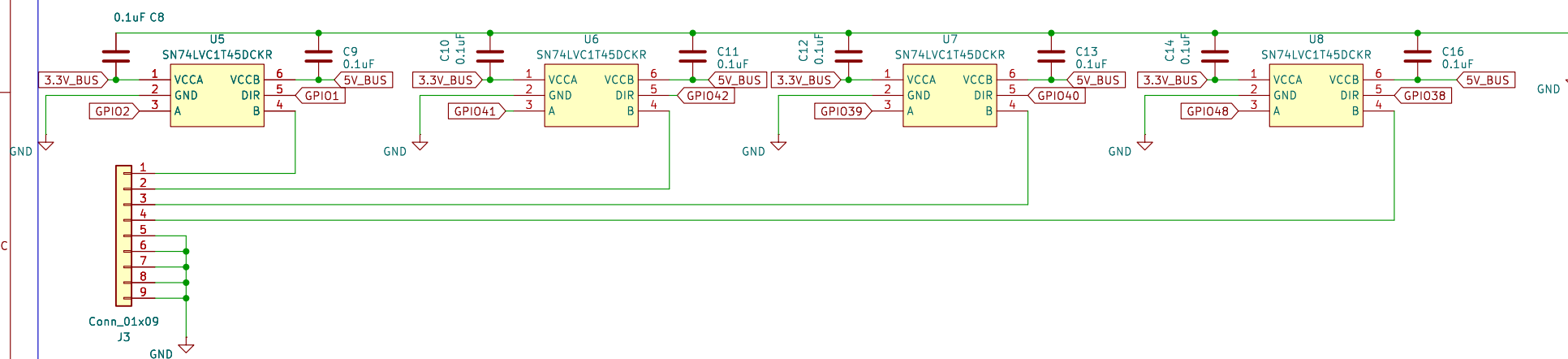


XT60 POWER

2S to 4s power input



**5V Logic IO Pins
Level Shifted
32mA**



Sheet: /Connectors/
File: Connectors.kicad_sch

Title:

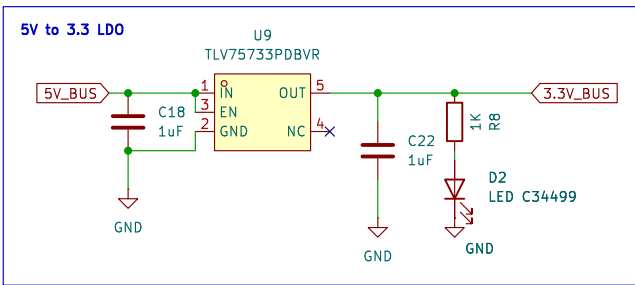
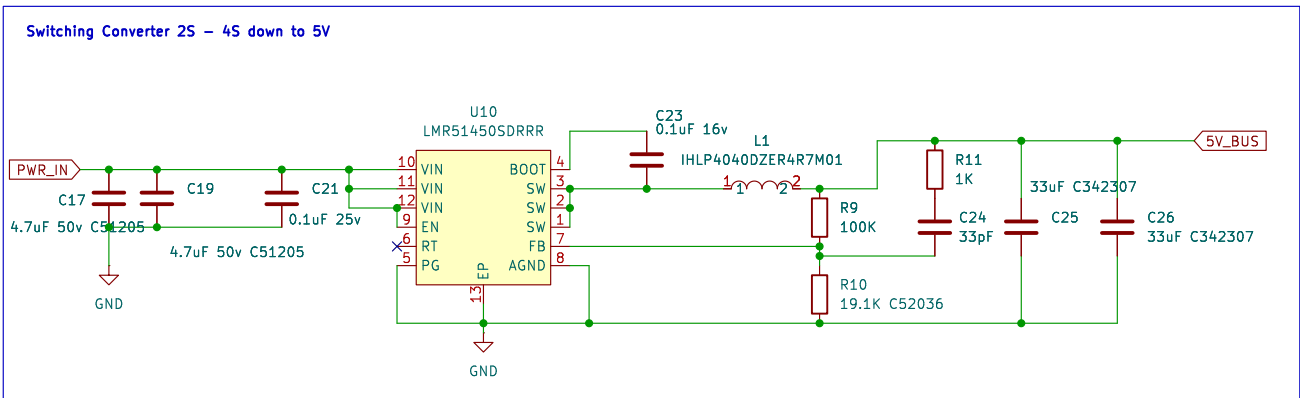
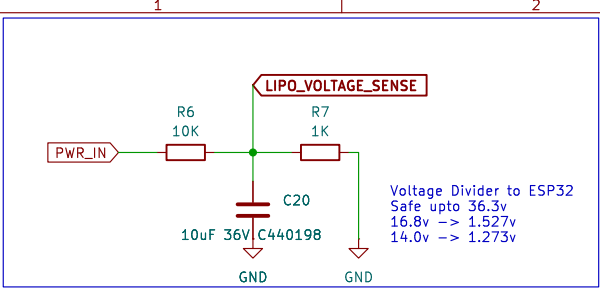
Size: A4

Date:

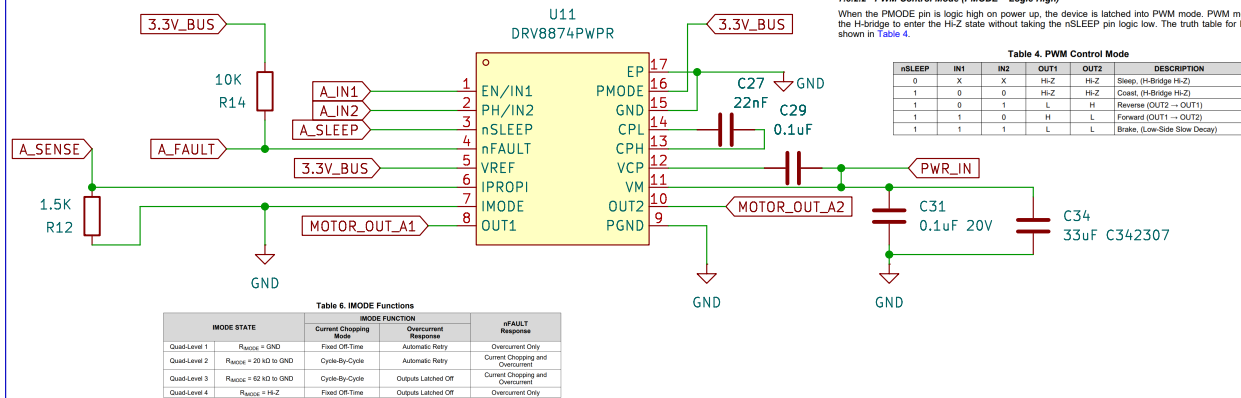
Size: A1	Date:
KiCad E.D.A. 9.0.2-rc1	

Rev:

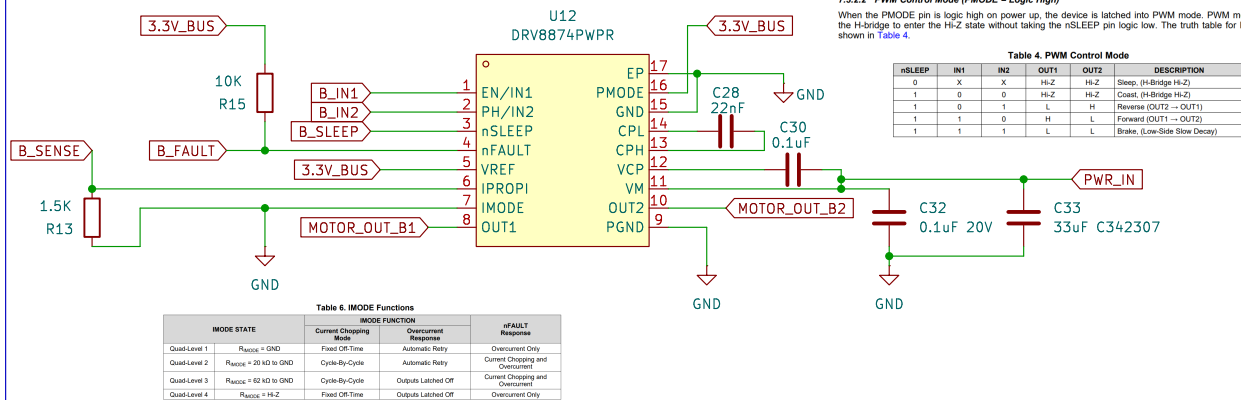
Id: 2/4



Left Motor Drive



Right Motor Drive



Sheet: /H Bridge Controllers/
File: H_Bridge.kicad_sch

Title:

Size: A4	Date:
KiCad E.D.A. 9.0.2-rc1	

Date:

Rev:
Id: 4/4